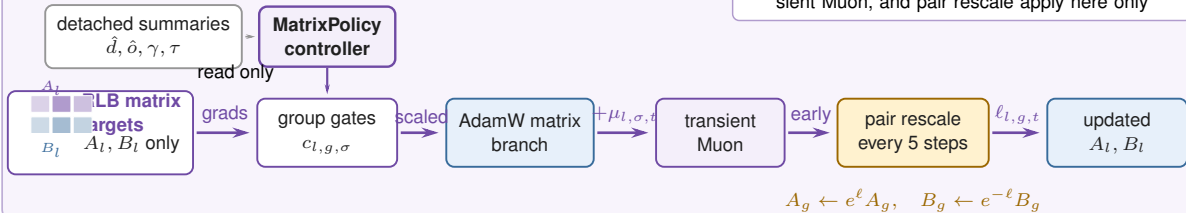


MatrixPolicy matrix lane: only RLB input/output matrices A_l, B_l

group gates, AdamW matrix branch, transient Muon, and pair rescale apply here only



AdamW-only bypass lane: rational coefficients and all other tensors



The gray lane is disjoint from MatrixPolicy updates; it may supply detached statistics but receives only AdamW steps.